

SOURABH PORWAL

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Senior Functional Safety Engineer · AV Safety, Standards & Regulation

Functional safety engineer with seven years in automotive and autonomous driving, across the safety areas this role touches: functional safety, operational safety, safety cases, and the regulation behind automated driving. At Vay, own safety for a remote-driving system on public roads, helped build the in-house safety function from scratch where no tele-driving regulation existed, and led the functional-safety side of an independent TÜV SÜD assessment. Track and apply the AV regulation and standards that shape the system (UNECE R 157, ISO 4804, ISO 26262, ISO 21448, the German tele-driving ordinance), analyse operational safety data such as minimum-risk-manoevre frequency, and work closely with OEM and partner safety teams.

CORE AREAS

AV Safety Standards & Regulation: UNECE R 157, ISO 4804, ISO 26262, ISO 21448, FMVSS, UL 4600, ISO/IEC TS 5083; German tele-driving ordinance; tracking and applying AV regulation in practice

Safety Engineering: Functional safety (full lifecycle), operational safety, safety cases; HARA, FMEA, FTA, FMEDA; risk assessment and mitigation

Operational Safety & Data: ODD definition; operational safety metrics and performance indicators (MRM frequency, reaction times); field-incident tracking

Partner & External Assessment: OEM and partner safety interface; B2B integration; supplier safety oversight; independent assessment (TÜV SÜD)

Building & Process: Safety functions and processes from scratch; requirements and traceability (JAMA, DOORS)

EXPERIENCE

Senior Functional Safety Engineer

09/2023 – Present

Vay Technologies GmbH · Berlin, Germany

- Track and apply the regulation and standards behind the remote-driving system (UNECE R 157, ISO 4804, the German tele-driving ordinance, FMVSS for US operations), keeping the system aligned as the rules evolve.
- Act as safety lead for integrating the remote-driving stack into a B2B partner's product, assessing and working directly with the partner's safety and engineering teams on architecture, integration, and safety deliveries.
- Analyse operational safety data, including how often vehicles trigger a minimum-risk manoeuvre, to track safety performance in the field.
- Own and author the safety case, and define the functional and technical safety architecture, including system-level redundancies.
- Lead the safety work for bringing AI into safety-critical driving functions.

Functional Safety Engineer II

06/2022 – 08/2023

Vay Technologies GmbH · Berlin, Germany

- Designed the system's fail-safe fallback behaviour (Minimum Risk Manoeuvre) end to end, from concept and systems through functional safety, applying ISO 26262, UNECE R 157, and ISO 4804.
- Led the functional-safety side of the independent TÜV SÜD assessment of the safety concept and risk analyses, contributing to Vay's first-in-Europe approval to operate on public roads without a safety driver.
- Performed safety analyses (FMEA, FTA) and supported verification and validation at system and vehicle level.

Functional Safety Engineer I

07/2021 – 06/2022

Vay Technologies GmbH · Berlin, Germany

- Helped establish Vay's in-house functional safety function and processes from scratch, deriving the safety framework from first principles where no teledriving regulation existed.
- Implemented safety requirements management and traceability (JAMA), and built the safety documentation and review process for certification readiness.

Functional Safety Engineer / Manager

01/2020 – 06/2021

BERNS Engineers GmbH · Gilching, Germany

- As functional safety manager on certain projects, owned the safety plan, oversaw supplier safety activities, and provided functional-safety releases for the OEM's internal development milestones.
- Prepared safety concepts and analyses (FSC, TSC, HARA, FTA) for OEM customers and worked directly with their architects on ASIL D ADAS features.

Functional Safety Engineer

06/2019 – 12/2019

AKKA Technologies · Ingolstadt, Germany

- Supported ISO 26262 development for an ASIL B system and managed vehicle configuration data.

Junior Executive

08/2013 – 09/2015

Fleetguard Filters Pvt. Ltd. · Pune, India

- Managed production part approval process (PPAP) activities across manufacturing suppliers, including process FMEA and process flow documentation; reduced component cost by ~25% through supplier negotiations.

EDUCATION

M.Sc. Mechanical Engineering

10/2015 – 04/2019

Rhine-Waal University of Applied Sciences · Kleve, Germany

B.E. Mechanical Engineering

08/2009 – 04/2013

Visvesvaraya Technological University · Belgaum, India

CERTIFICATION

CertX – ISO 26262:2018 Functional Safety Red Belt (03/2021)

PERSONAL DETAILS

Nationality: German **Mobility:** Open to relocation

Languages: English (fluent), German (B1), Hindi (native)